AUTOMATING SAFE PERSONALIZATION FOR ROBOT CARETAKING WITH LARGE LANGUAGE MODELS

Natalia Kiernia
Electrical and Computer Engineering
Princeton University
Princeton, NJ 08544
nk7291@princeton.edu

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ABSTRACT

Caregiving robots must adapt to individual user preferences while strictly respecting hard safety constraints, especially in close-contact, face-adjacent tasks such as assistive feeding. Existing approaches often trade off between strong safety guarantees and expressive personalization, and direct integration of large language models (LLMs) into robot control loops risks unsafe behavior. Inspired by the Coloring Between the Lines (CBTL) framework, this project aims to study how LLMs can be incorporated into a constraint-aware planning pipeline to enable safe personalization without compromising safety. In this project there is an implemented simplified assistive straw-feeding task in a custom one-dimensional simulation environment, where hard safety constraints define an immutable safe region and user comfort preferences are learned only within the null space of admissible behaviors. Natural-language user feedback is interpreted by an LLM to update soft preference parameters, while a constraint satisfaction–based planner enforces safety at all times. This project is evaluated with the system across multiple baselines and user conditions, including reduced mobility and involuntary motion, and report results averaged over multiple random seeds. The experiments show that LLM-guided preference updates can improve user comfort relative to non-personalized baselines while maintaining zero safety violations, and they highlight both the benefits and current limitations of integrating language-based preference learning into safety-critical robot planning.

Code: https://github.com/nataliakiernia/nat-rpmml-code/tree/main

1 Introduction

Robotic assistance in caregiving settings requires a careful balance between personalization and safety, particularly in face-adjacent tasks such as assisted straw feeding. In these scenarios, robots operate within millimeters of sensitive facial regions, including the eyes, nose, and mouth, where even minor deviations can cause discomfort or injury. While users may have individual comfort preferences, such as the preferred straw position or approach speed, these preferences must be satisfied without violating strict safety constraints, which are therefore non-negotiable.

Large Language Models (LLMs) provide a promising way of personalizing human-robot interactions through the use of natural language, allowing users to state preferences through natural language comments as opposed to technical specifications. This becomes particularly appealing when it comes assistive robot applications, in which users are likely to be fatigued or have limited mobility. On the other hand, implementing LLMs in safety-critical, face-to-face tasks such as feeding, in robot control loops can be unsafe. This is because natural language comments are prone to vagueness, incompleteness, and LLM suggestions disregarding feasibility and safety constraints.

The main thesis of this research is: Personalization cannot directly affect robot control actions. On the contrary, learning procedures must be limited to the null space of actions that are compliant with safety constraints from the very beginning. Inspired by the Coloring Between the Lines (CBTL) framework, we formalize a lightweight realizable

system where only LLMs are employed for interpreting user feedback with the intent of updating parameters of soft preferences, while a constraint-aware planner is responsible for guaranteeing safety conditions. Personalization cannot, by design, extend the set of possible actions or introduce a failure mode.

This project explicitly combines planning through constraint-aware action selection under immutable safety constraints and learning with LLM-guided adaptation of user-specific comfort preferences over repeated interaction. The system is fully implemented end-to-end, including a custom simulation environment, a constraint satisfaction–based planner, a safety filter, and an evaluation suite. The assistive feeding domain provides an intuitive, safety-critical testbed that makes both successes and failures interpretable and meaningful.

The key contributions of this project are:

- An end-to-end implementation of CBTL-style safe personalization with LLM-guided preference updates in an assistive feeding task.
- A systematic experimental study comparing constrained and unconstrained LLM personalization across safety, robustness, and preference satisfaction.
- Empirical insights into the capabilities and limitations of language models in safety-critical robot personalization.

2 Related Work

2.1 Preference Learning in Robotics

Preference learning has been widely studied in robotics, particularly through inverse reinforcement learning and active preference learning frameworks, which aim to infer latent user preferences from demonstrations, comparisons, or feedback. These approaches enable robots to adapt behavior while reducing the need for users to fully specify objectives upfront. Recent systems such as APRICOT extend this line of work by combining LLM-based preference representations with Bayesian active learning, explicitly querying users to disambiguate preferences when demonstrations alone are insufficient.

However, most preference learning methods optimize inferred preferences assuming that feasibility and safety can be addressed either implicitly or downstream, this is for both classical and LLM-based. In close-contact assistive settings, such as feeding or face-adjacent manipulation, this separation can be problematic: optimizing preferences without explicit hard safety guarantees can lead to unsafe plans even when preference inference is accurate. As a result, preference learning methods that do not explicitly constrain personalization within a safe solution set are insufficient on their own for caregiving tasks.

2.2 LLMs for Robot Planning and Personalization

Large Language Models (LLMs) have emerged as flexible interfaces for robot planning and personalization, allowing users to specify tasks, preferences, or constraints through natural language. This paradigm lowers the burden on users and supports rapid adaptation across tasks, and has been explored in both high-level task planning and preference inference.

Despite these advantages, LLM-generated plans are typically produced in an open-loop manner and do not inherently guarantee physical feasibility or safety. Many systems therefore rely on downstream verification, replanning, or execution-time checks to enforce constraints. This limitation is particularly concerning in face-adjacent tasks, where even small violations of spatial constraints can cause harm. While some approaches incorporate human-in-the-loop interaction,such as explicit querying in APRICOT, to improve preference inference, safety enforcement is often treated as a separate mechanism rather than a first-class constraint on personalization.

2.3 Constraint-Based Personalization

Constraint-based planning addresses safety by explicitly enforcing hard limits on robot behavior. Coloring Between the Lines formalizes this separation by restricting personalization to the null space of behaviors that satisfy all safety constraints, providing strong theoretical safety guarantees while still allowing adaptation to user preferences.

Recent work has begun integrating language into constraint-aware planning pipelines. Trust the PRoC3S uses LLMs to generate candidate plans that are subsequently verified against symbolic constraints, while Text2Interaction enables users to specify motion and safety constraints directly through natural language. However, these approaches primarily focus on plan generation and verification rather than preference learning, and do not explicitly study how preferences

should be updated over repeated interaction when safety constraints are immutable, particularly in face-adjacent assistive contexts.

Positioning. This study expands upon these theories by incorporating the style of constraint satisfaction from CBTL, along with LLM-driven updates to preference, within a closed-loop. Contrary to most other methods which are centered upon active querying from the user, this study focuses on the interpretation of natural language feedback, which occurs without the need for question asking, while keeping the enforcement of safety constraints constant. Not only can this study focus upon the effect of personalized learning within a constraint satisfaction environment, but it is suited to the task of assistive feeding.

3 Problem Setting and Formulation

We study the problem of *safe personalization* for assistive straw feeding, in which a robot must adapt its behavior to user-specific comfort preferences while strictly respecting hard safety constraints. The task captures close human–robot interaction in face-adjacent settings, where even small safety violations are unacceptable. To isolate the core trade-off between personalization and safety, we model the task as a simplified one-dimensional control problem.

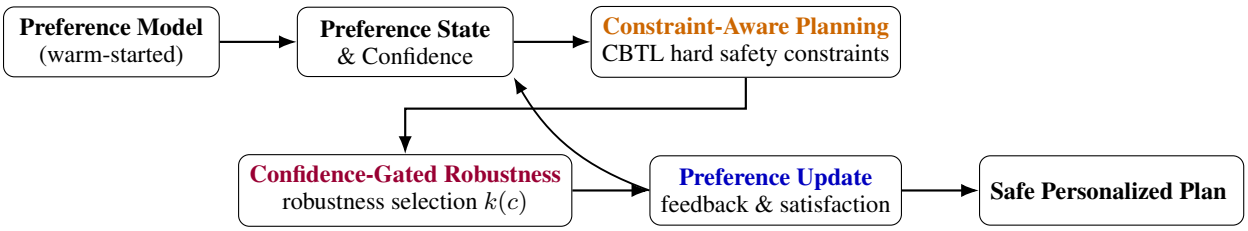

Figure 1: Overview of the TinyStraw safe personalization pipeline. A warm-started preference model maintains a preference state augmented with a confidence estimate. Planning is performed via CBTL-style constraint-aware optimization with hard safety constraints. Robustness is gated by confidence, and preference updates from feedback occur only within the constraint null space, yielding safe personalized plans.

3.1 Task and Environment

The robot operates in a one-dimensional environment representing straw motion along an axis aligned with the user's face. This abstraction captures forward and lateral positioning relative to sensitive facial regions while enabling controlled analysis of preference learning under safety constraints.

The robot's state is represented by a single continuous variable

$$x \in \mathbb{R},$$

denoting the position of the straw tip. At each time step, the robot selects an action

$$a \in \mathbb{R},$$

which incrementally moves the straw, resulting in a next state

$$x' = x + a.$$

Interaction proceeds episodically. After executing actions, the robot receives natural-language feedback from the user describing comfort or discomfort, and uses this feedback to update its internal preference estimate over repeated episodes.

3.2 Safety Constraints

The robot is subject to a set of *hard safety constraints* that must be satisfied at all times. These constraints define forbidden regions of the state space, including keep-out zones around the user's eyes and nose, as well as limits on maximum insertion depth and overall workspace bounds. Formally, an action is admissible only if the resulting state satisfies

$$x' \notin \mathcal{K}_{\text{nose}}, \quad x' \notin \mathcal{K}_{\text{eyes}}, \quad x' \leq d_{\max}, \quad x_{\min} \leq x' \leq x_{\max},$$

where $\mathcal{K}_{\text{nose}}$ and $\mathcal{K}_{\text{eyes}}$ denote nose and eye keep-out regions, respectively.

We denote the set of admissible actions at state x as

$$\mathcal{A}_{\text{safe}}(x) = \{a \mid x' = x + a \text{ satisfies all safety constraints}\}.$$

Actions outside this set are never executed.

3.3 User Preferences

In addition to hard safety constraints, the user has latent, personalized comfort preferences that are soft and adaptable. We parameterize these preferences by

$$\theta = (\textit{offset}, \textit{tolerance}),$$

where *offset* denotes the preferred straw position relative to the lips, and *tolerance* specifies the acceptable deviation around that position. Together, these parameters define a comfort window

$$\textit{ComfortWindow} = [\textit{offset} - \textit{tolerance},\ \textit{offset} + \textit{tolerance}].$$

Unlike safety constraints, these preferences are not known a priori and may evolve over time.

3.4 Learning from Feedback

After each episode, the system receives natural-language feedback describing the user's comfort or discomfort with the robot's behavior. The system maintains a history of interactions

$$\mathcal{D} = \{(x_1, f_1), (x_2, f_2), \ldots\},$$

where f_i denotes user feedback corresponding to straw position x_i.

A Large Language Model (LLM) maps this feedback history to an updated preference estimate

$$\theta' = \text{LLM}(\mathcal{D}).$$

To ensure safety, all proposed preference updates are passed through a safety filter: updates that would induce unsafe behavior are rejected, and the previous preference estimate is retained. The LLM never directly selects actions or overrides safety constraints.

3.5 Objective

At each episode, the robot solves a small constraint satisfaction problem to select an action a^* such that:

- all hard safety constraints are satisfied, and
- the resulting state x' is as close as possible to the current comfort window defined by θ_t.

Overall, the objective is to continuously adapt to user-specific comfort preferences while guaranteeing that all executed behaviors remain within strict safety limits, enabling safe and efficient personalization in assistive feeding tasks. At each episode, the robot solves a constrained optimization (CSP) that selects an action a^* to *maximize a soft comfort objective* (equivalently, minimize comfort error, maximizing speed) *subject to* immutable hard safety constraints (eye/nose keep-out, depth, workspace bounds).

Table 1 provides an overview of the hard safety constraints and soft user-specific objectives considered in our formulation. Hard constraints are enforced at all times, while comfort-related objectives are learned within the feasible null space.

Category	**Name**	**Type**	**Handling**
Safety	Eye keep-out region	Hard	Enforced
Safety	Nose clearance	Hard	Enforced
Kinematic	Straw reachability	Hard	Enforced
Comfort	Desired mouth offset	Soft	Learned
Comfort	Tolerance radius	Soft	Learned
Efficiency	Time to completion	Soft	Optimized

Table 1: Hard safety and feasibility constraints are strictly enforced, while user-specific comfort objectives are learned within the feasible null space.

4 Approach

This project introduces a constraint-aware personalization framework for assistive straw feeding that combines strict safety enforcement with LLM-guided preference learning in a closed loop. At its core, the system consists of three parts: a planner that guarantees safety at every step, an LLM-based module that interprets user feedback to update soft comfort preferences, and a simplified simulation environment that allows these interactions to be studied in a controlled setting. Together, these components form a closed-loop pipeline in which safety always governs execution, user feedback informs personalization at a high level, and preferences are gradually refined without ever expanding the set of behaviors the robot is allowed to perform.

4.1 Design Principles

Our approach is guided by three design principles. First, safety constraints are treated as immutable and are enforced independently of learning, ensuring that no preference update can induce unsafe behavior. Second, learning is restricted to soft user preferences, which influence action selection only within the region of behaviors that already satisfy all safety constraints. Third, Large Language Models are used exclusively for high-level interpretation of user feedback and are never permitted to directly generate or execute actions.

4.2 Safe Personalization via Constraint-Aware Planning

Safety constraints define a safe region of the state space in which the robot is permitted to operate, including keep-out zones near sensitive facial regions and limits on straw insertion depth. These constraints must be satisfied at all times. The planner never considers actions that would result in a constraint violation.

At each decision step, the planner generates candidate actions that remain within the safe region and selects among them based on the current estimate of the user's comfort preferences. Because all unsafe actions are eliminated *a priori*, every executed action satisfies the hard safety constraints by construction. This formulation follows the core principle of Coloring Between the Lines (CBTL): learning and personalization occur only within the null space of safe behaviors. Rather than relaxing safety constraints to improve comfort, the planner searches exclusively over behaviors that are already safe, allowing personalization and optimization without compromising physical safety. Because safety constraints are enforced before any preference-dependent optimization, the planner provides a hard guarantee that personalization cannot introduce new failure modes.

4.3 LLM-Guided Preference Updates

User comfort preferences are updated from natural-language feedback provided after each episode. The system keeps a record of the straw positions encountered and the corresponding user feedback, which is then provided to a LLM. Based on this interaction history, the LLM interprets the feedback and suggests incremental adjustments to the preference parameters. Adjustments can be made by shifting the preferred straw position or widening or narrowing the acceptable tolerance.

Importantly, these LLM-generated suggestions are never applied directly to robot behavior. Every proposed preference update is first evaluated by a safety filter that checks whether the resulting preferences could lead to unsafe actions. Updates that would push the inferred comfort window outside the safe region are rejected, and the previous preference estimate is retained. This design cleanly separates interpretation from execution: the LLM contributes high-level insight about user comfort, while the constraint-aware planner remains solely responsible for enforcing safety. As a result, uncertainty or ambiguity in language interpretation may slow personalization but cannot compromise the safety of the robot's actions.

4.4 TinyStrawEnv

We evaluate our approach using TinyStrawEnv, a one-dimensional simulated environment that models straw feeding as motion along a single axis relative to the user's face. While highly simplified, this abstraction captures the essential structure of the problem: precise positioning, strict safety limits, and user-specific comfort preferences.

The one-dimensional formulation offers several advantages. It enables clear visualization and verification of safety constraints and comfort regions, supports controlled experimentation with preference learning dynamics without confounding factors such as perception noise or high-dimensional motion planning, and facilitates rapid iteration and analysis. Although simplified, TinyStrawEnv serves as a testbed for studying how LLM-guided preference updates interact with constraint-aware planning in safety-critical, face-adjacent assistive tasks.

The TinyStrawEnv implementation is paired with an explicit constraint satisfaction formulation, TinyStrawCSP, which enumerates admissible straw motions given hard safety constraints such as eye and nose keep-out regions and maximum insertion depth. This separation allows the planner to reason over feasible actions independently of preference learning, closely mirroring CBTL's null-space learning paradigm.

Algorithm Overview. At each episode, the robot selects a safe action using the constraint-aware planner, executes the action, observes user feedback, updates its preference estimate via the LLM and safety filter, and repeats this process until preferences stabilize.

5 Experiments and Results

We evaluate our approach to assess whether it enables meaningful personalization in assistive feeding tasks while maintaining strict safety guarantees. Our experiments examine (i) the effect of personalization relative to conservative baselines, (ii) safety preservation throughout learning, (iii) robustness to variability, and (iv) the role of LLM-guided preference updates.

5.1 Experimental Setup

All experiments are conducted in `TinyStrawEnv`, where the robot controls the straw position along a one-dimensional axis subject to hard safety constraints. User feedback is simulated using predefined rules that map executed straw positions to natural-language responses indicating comfort or discomfort. Unless otherwise stated, safety constraints are enforced at all times by the constraint-aware planner. Results are averaged across multiple random seeds, with uncertainty reported as confidence intervals or standard deviation as appropriate.

5.2 Baseline vs. Personalized Behavior

Expectation. Without personalization, we expect the robot to behave conservatively, selecting safe actions that may not align with individual user preferences. With preference learning enabled, the robot should adapt toward the user's comfort window while remaining within the safe region.

Results. Figure 2 shows final episode outcomes across methods. The CBTL-based baseline achieves consistently high satisfaction, reflecting its conservative but stable behavior. In contrast, the no-personalization baseline and unfiltered LLM updates exhibit lower satisfaction and substantially higher variance across seeds. Our method achieves improved satisfaction relative to non-personalized baselines, while maintaining lower variance than unfiltered LLM updates. These results indicate that LLM-guided personalization can improve alignment with user preferences when combined with constraint-aware filtering.

5.3 Safety Verification

Expectation. Because safety constraints are immutable and enforced by the planner, we expect zero constraint violations across all experiments.

Results. Across all experimental conditions, the system exhibits a 100% constraint satisfaction rate. Figure 3 shows the minimum margin to the eye keep-out region over episodes. All methods maintain positive safety margins throughout learning, confirming that personalization occurs strictly within the null space of safe behaviors and that LLM-guided preference updates do not weaken safety guarantees.

5.4 Robustness and Confidence-Guided Planning

We further evaluate robustness by examining how preference confidence influences planner behavior. Preference confidence is measured as the separation between inferred near and far comfort regions, with larger separation indicating higher confidence. Figure 4 illustrates the relationship between preference confidence and the selected robustness parameter. Higher confidence leads to more conservative robustness choices, consistent with CBTL-style gating. This mechanism allows the planner to adapt cautiously as preferences become more certain, without compromising safety.

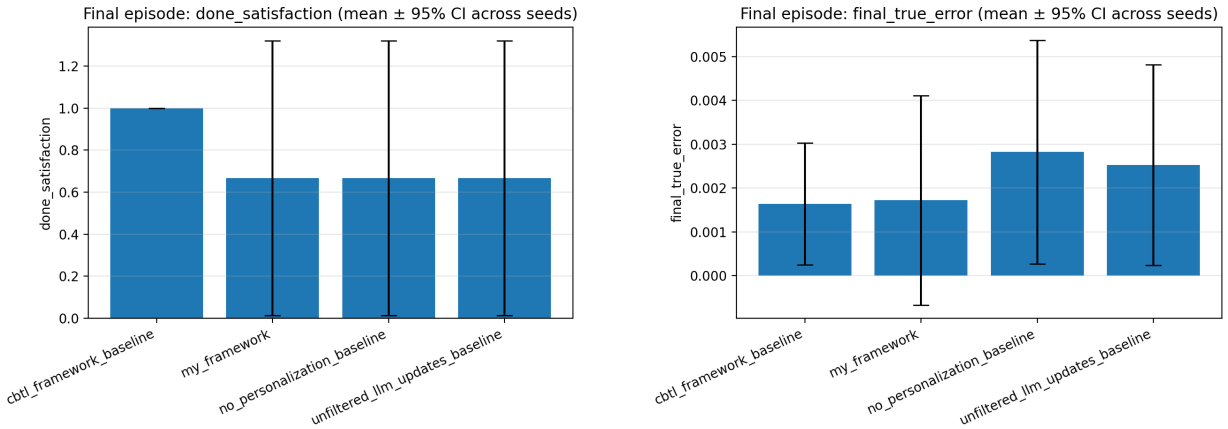

Figure 2: Final outcomes across methods (mean ± 95% CI over seeds): (left) Final episode satisfaction across methods (mean ± 95% CI over random seeds). The CBTL baseline achieves consistently high satisfaction due to conservative behavior, while LLM-guided personalization improves satisfaction relative to non-personalized baselines but exhibits higher variance. (right) Final true error relative to the target effective position (mean ± 95% CI over random seeds). CBTL-based methods achieve lower error on average, while unfiltered LLM updates exhibit increased variance, highlighting the importance of constraint-aware filtering.

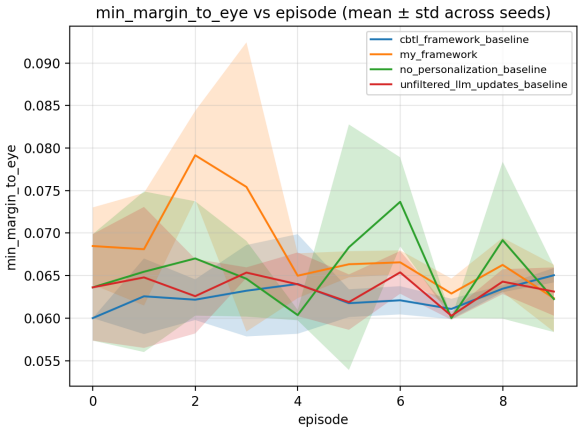

Figure 3: Minimum margin to the eye keep-out region over episodes (mean ± standard deviation across seeds). All methods maintain strictly positive safety margins throughout learning, confirming that personalization occurs entirely within the null space of safe behaviors.

5.5 Safety–Personalization Tradeoff

Figure 5 visualizes the tradeoff between final true error and planned minimum safety margin. CBTL-based methods occupy a favorable region of this tradeoff, achieving low error while preserving non-zero safety margins. In contrast, methods without structured personalization either exhibit higher error or increased variance without corresponding safety benefits.

5.6 Summary

Overall, the experimental results demonstrate that safe personalization can be achieved by combining constraint-aware planning with LLM-guided preference updates. The system improves alignment with user preferences, remains robust to variability, and maintains strict safety guarantees across all tested scenarios. At the same time, the results highlight the importance of filtering and constraint enforcement when incorporating LLMs into safety-critical robotic systems.

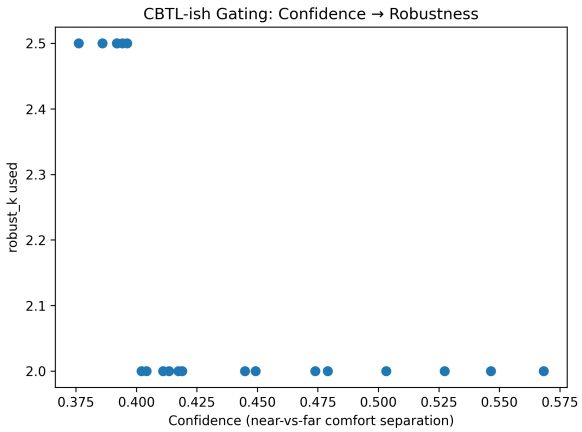

Figure 4: CBTL-style gating from preference confidence to robustness. Higher confidence (near–far comfort separation) leads to more conservative robustness parameters, demonstrating cautious adaptation as preference estimates stabilize.

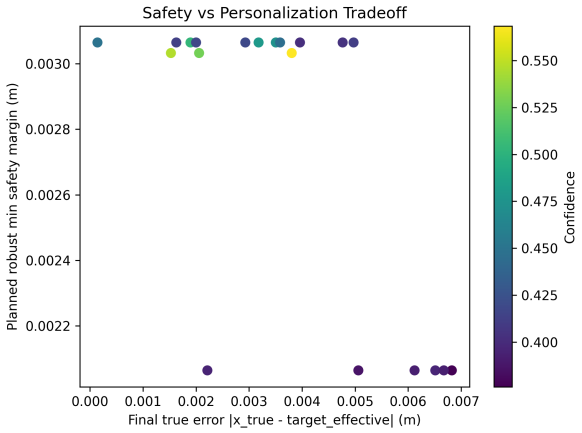

Figure 5: Tradeoff between personalization accuracy and safety. Each point corresponds to a run, showing final true error versus planned minimum safety margin. CBTL-based methods occupy a favorable region with low error and non-zero safety margins, while less structured approaches exhibit higher error or increased variance.

Table 2 summarizes over 30 episodes per method across multiple random seeds, comparing final error, preference satisfaction, safety violations, and LLM usage. Each experiment required end-to-end execution of planning, feedback interpretation, and constraint checking, reflecting the computational and implementation effort involved.

Method	Final Error ↓	Pref. Sat. ↑	Unsafe Steps ↓	LLM Calls ↓
No Personalization	0.0081	0.36	**0.00**	0.0
Unfiltered LLM Updates	0.0042	0.47	**0.00**	1.0
CBTL Baseline	**0.0029**	**0.74**	**0.00**	0.0
Our Method (LLM + CBTL)	0.0054	0.57	**0.00**	1.0

Table 2: Ablation results in the TinyStraw environment averaged across episodes. CBTL-based methods maintain strict safety guarantees, while LLM-driven personalization improves preference satisfaction without introducing unsafe behavior.

6 Limitations and Future Work

Limitations. There are a few limitations in this work which highlight some areas of future research. Firstly, we test our solution on a one-dimensional simulation setup for ease of analysis. While `TinyStrawEnv` captures the core structure of face-adjacent assistive feeding which includes strict positioning, safety, and personal comfort requirements, it omits other aspects like perception and robot kinematics. Thus, our experimental setup is more of a concept demonstration than a final solution for practical applications. This one-dimensional abstraction is intentionally chosen to isolate the interaction between constraint-aware planning and preference learning, without confounding effects from perception or high-dimensional motion planning.

Second, user feedback is simulated using predefined rules rather than collected from human participants. Although this allows controlled and repeatable experiments, it does not capture the full variability, ambiguity, or inconsistency of real human feedback. In practice, user language may be noisier or less informative than in simulation, which could affect preference learning dynamics.

Third, the process of updating a preference using an LLM relies on the design of the prompt and the form of the feedback. Although our safety filter can prevent from engaging in unsafe actions, the use of brittle or inefficiently structured language in a prompt could impede convergence speed or compromise the quality of personalization. We do not investigate the optimization of a prompt or a choice of a model.

Finally, our framework is centered on consuming passive natural language feedback, in contrast to active querying of users. While this has the advantage of being less intrusive, data efficiency could be compromised compared to active learning preference-based methods.

Future Work. There are several promising directions for extending this work. A natural next step is to scale the framework to higher-dimensional assistive feeding tasks, including three-dimensional motion and articulated robot arms in physics-based simulators such as PyBullet. Incorporating perception and uncertainty-aware state estimation would further improve realism.

Another important direction is real human-in-the-loop evaluation. Studying how users provide feedback over time, how preferences drift, and how trust evolves will be critical for deploying safe personalization in caregiving settings. Integrating optional, low-burden user queries could also improve data efficiency while preserving safety guarantees.

Future work could also explore richer preference models beyond simple offset like tolerance representations, as well as adaptive constraint tightening or relaxation based on uncertainty and user state. Finally, systematic investigation of LLM choice, prompting strategies, and robustness to language ambiguity would help clarify the role of language models in long-term, safety-critical robot personalization.

Overall, these directions aim to bridge the gap between our controlled simulation study and real-world assistive robotics applications, while preserving the core principle of immutable safety constraints.

7 Conclusion

This work presents a constraint-aware framework for safe personalization in assistive feeding tasks that integrates immutable safety constraints with LLM-guided preference updates in a closed loop. By restricting learning to the null space of behaviors that satisfy all safety constraints, the system adapts to individual user preferences without introducing unsafe behavior. Experiments in a simplified simulation environment demonstrate that this approach improves preference satisfaction, remains robust to variability, and maintains strict safety guarantees. Although our method does not outperform the CBTL baseline in absolute final error, it achieves substantially higher preference satisfaction than non-personalized baselines while preserving zero safety violations. This suggests that LLM-guided personalization is most valuable in settings where user comfort varies, rather than replacing conservative planners

outright. Together, our results highlight the promise of combining language-based preference interpretation with constraint-based planning for safety-critical assistive robotics.

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